

# FITTING A GEV IN JULIA

An aerial photograph showing a residential neighborhood in Julia, Mississippi, completely inundated with floodwater. The water is a murky brown color, reaching the roofs of many houses and surrounding the trees. The houses have various roof colors, including grey, brown, and red. The trees are mostly green, though some appear to be partially submerged. A central road runs vertically through the middle of the image, also flooded. In the upper right corner, a green field and a body of water are visible, contrasting with the flooded area.

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CEVE 543

2026-09-16

## **Port Pirie data**

Maximum likelihood by hand

L-moment estimation

Gumbel vs GEV

Diagnostics

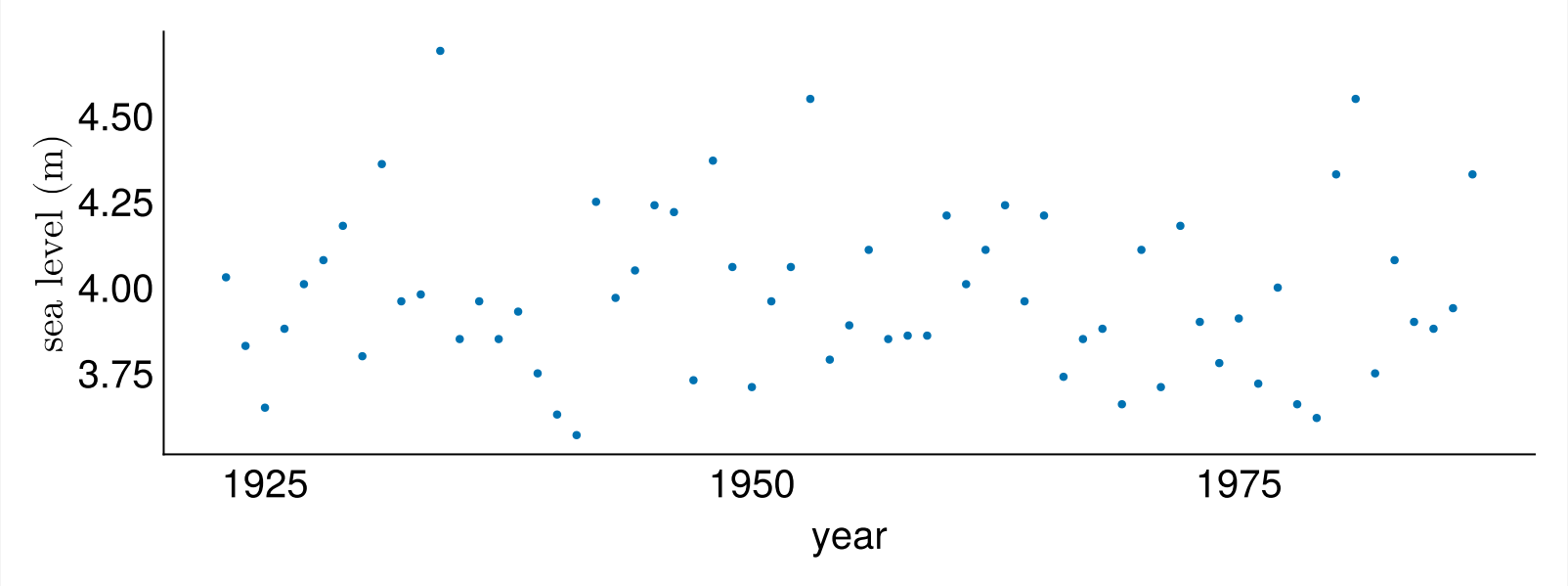
Bayesian inference

# ANNUAL MAXIMUM SEA LEVELS AT PORT PIRIE

65 years of annual maximum sea level (metres) at Port Pirie, South Australia, 1923–1987 (Coles, 2001, §3.4).

```
1 portpirie = CSV.read(joinpath(here, "portpirie.csv"), DataFrame)
2 y = portpirie.SeaLevel
3 fig = Figure(; size=(800, 300))
4 ax = Axis(fig[1, 1]; xlabel="year", ylabel=L"\text{sea level (m)}")
5 scatter!(ax, portpirie.Year, y; color=colors[1], markersize=6)
6 fig
```

# ANNUAL MAXIMUM SEA LEVELS AT PORT PIRIE

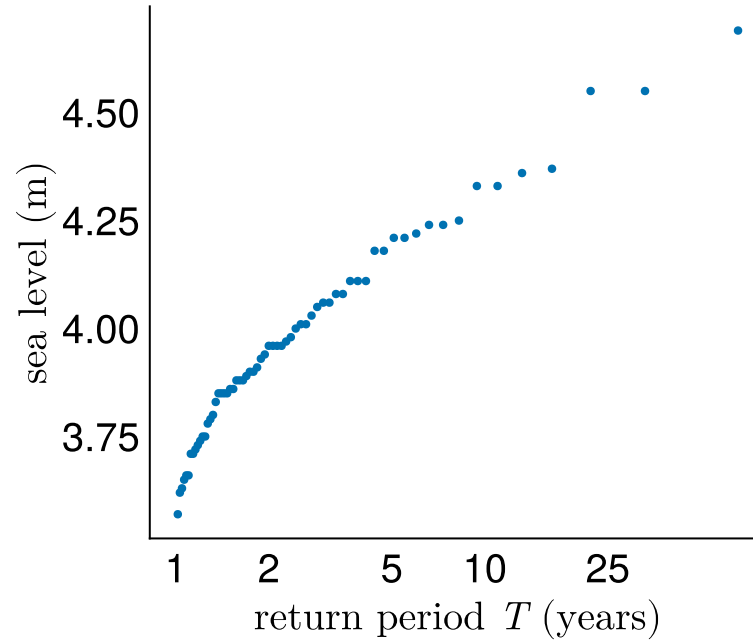
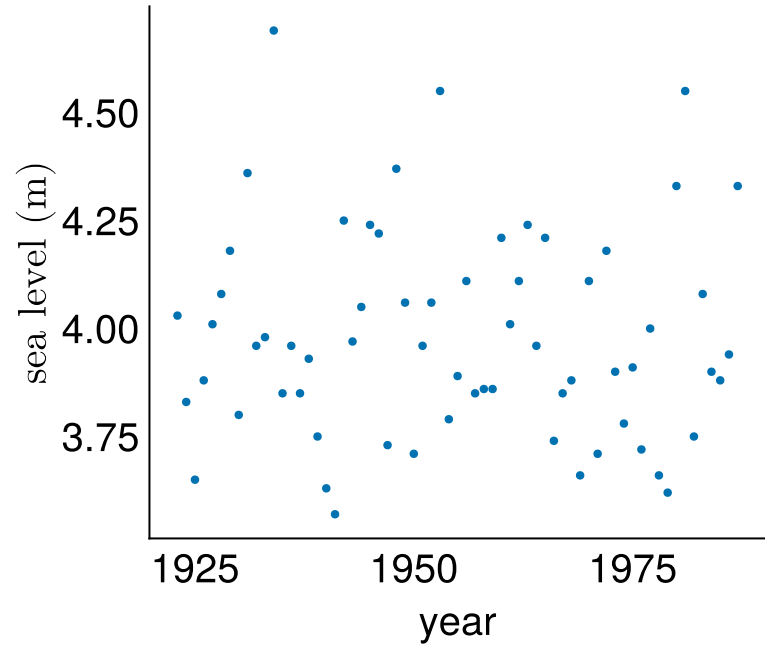


# THE DATA ON A RETURN-PERIOD SCALE

Rank the data and assign Weibull plotting positions  $\hat{T}_i = (n + 1)/i$ .

```
1 sorted, p_hat, T_hat = plotting_positions(y)
2 fig = Figure(; size=(800, 350))
3 ax1 = Axis(fig[1, 1]; xlabel="year", ylabel=L"\text{sea level (m)}")
4 scatter!(ax1, portpirie.Year, y; color=colors[1], markersize=6)
5 ax2 = Axis(fig[1, 2]; ylabel=L"\text{sea level (m)}")
6 linkyaxes!(ax1, ax2)
7 scatter!(ax2, T_hat, sorted; color=colors[1], markersize=6)
8 return_period_axis!(ax2)
9 fig
```

# THE DATA ON A RETURN-PERIOD SCALE



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# THE GEV LOG-DENSITY

From the board: the GEV has three parameters  $(\mu, \sigma, \xi)$  and its log-density is

$$\log f(z \mid \mu, \sigma, \xi) = -\log \sigma - (1 + 1/\xi) \log t - t^{-1/\xi}$$

where  $t = 1 + \xi(z - \mu)/\sigma > 0$ .

# THE NEGATIVE LOG-LIKELIHOOD

MLE finds the parameter vector that maximises the sum of log-densities, or equivalently minimises the negative.

```
1 function gev_nll( $\theta$ , y)
2      $\mu$ , log $\sigma$ ,  $\xi$  =  $\theta$ 
3      $\sigma$  = exp(log $\sigma$ )           # optimise in log  $\sigma$  so scale stays positive
4     d = GeneralizedExtremeValue( $\mu$ ,  $\sigma$ ,  $\xi$ )
5     nll = 0.0
6     for  $y_i$  in y
7         lp = logpdf(d,  $y_i$ )
8         isfinite(lp) || return 1e10 # penalty for out-of-support proposals
9         nll -= lp
10    end
11    return nll
12 end
```

# OPTIMISING WITH NELDER-MEAD

```
1   $\theta_0$  = [mean(y), log(std(y)), 0.1]      # initial guess: sample moments, slight positive
shape
2  result = optimize(theta -> gev_nll(theta, y),  $\theta_0$ , NelderMead())
3   $\hat{\mu}$ , log $\hat{\sigma}$ ,  $\hat{\xi}$  = Optim.minimizer(result)
4   $\hat{\sigma}$  = exp(log $\hat{\sigma}$ )
5  @printf(" $\hat{\mu}$  = %.3f,  $\hat{\sigma}$  = %.3f,  $\hat{\xi}$  = %.3f",  $\hat{\mu}$ ,  $\hat{\sigma}$ ,  $\hat{\xi}$ )
```

```
 $\hat{\mu}$  = 3.875,  $\hat{\sigma}$  = 0.198,  $\hat{\xi}$  = -0.050
```

# THE CONVENIENCE WRAPPER

`CEVE543Utils.gevfit` wraps the same optimization with support for covariates.

```
1 fit_gev = gevfit(y)
2 d_gev = only(getdistribution(fit_gev))
3 @printf("gevfit:  $\hat{\mu}$  = %.3f,  $\hat{\sigma}$  = %.3f,  $\hat{\xi}$  = %.3f", d_gev. $\mu$ , d_gev. $\sigma$ , d_gev. $\xi$ )
```

```
gevfit:  $\hat{\mu}$  = 3.875,  $\hat{\sigma}$  = 0.198,  $\hat{\xi}$  = -0.050
```

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# L-MOMENTS

MLE requires an optimizer and can be sensitive to starting values or small samples. L-moments are linear combinations of order statistics that give closed-form parameter estimates: no optimizer, always finite, and often more stable in small samples.

They are estimated from probability weighted moments  $b_r = \frac{1}{n} \sum_{i=1}^n y_{(i)} \binom{i-1}{r} / \binom{n-1}{r}$ .

The first three L-moments:

$$\ell_1 = b_0 \quad \ell_2 = 2b_1 - b_0 \quad \ell_3 = 6b_2 - 6b_1 + b_0$$

$\ell_1$  is the mean,  $\ell_2$  measures spread, and the L-skewness ratio  $\tau_3 = \ell_3/\ell_2$  measures asymmetry.

# FROM L-MOMENTS TO GEV PARAMETERS

The shape follows from the PWM ratios through the polynomial approximation of Hosking, Wallis & Wood (1985):

$$c = \frac{2b_1 - b_0}{3b_2 - b_0} - \frac{\ln 2}{\ln 3} \quad k = 7.859c + 2.955c^2 \quad \xi = -k$$

Then scale and location follow in closed form:

$$\sigma = \frac{k(2b_1 - b_0)}{(1 - 2^{-k})\Gamma(1 + k)} \quad \mu = b_0 - \frac{\sigma}{k}(1 - \Gamma(1 + k))$$

```
1 d_lmom = gevfit_lmom(y)
2 @printf("L-moments:  $\hat{\mu}$  = %.3f,  $\hat{\sigma}$  = %.3f,  $\hat{\xi}$  = %.3f", d_lmom.mu, d_lmom.sigma, d_lmom.xi)
```

# FROM L-MOMENTS TO GEV PARAMETERS

L-moments:  $\hat{\mu} = 3.873$ ,  $\hat{\sigma} = 0.203$ ,  $\hat{\xi} = -0.051$

# TWO ESTIMATORS, TWO ANSWERS

MLE:  $\mu = 3.875, \sigma = 0.198, \xi = -0.0501$

L-moment:  $\mu = 3.873, \sigma = 0.203, \xi = -0.0515$

Different estimators give different answers on the same data, especially for  $\xi$ . The shape parameter is the hardest to estimate and the most consequential for the tail.

Port Pirie data

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**Gumbel vs GEV**

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# DOES THE SHAPE PARAMETER MATTER?

The Gumbel is the GEV with  $\xi = 0$  forced: the tail is exponential by assumption.

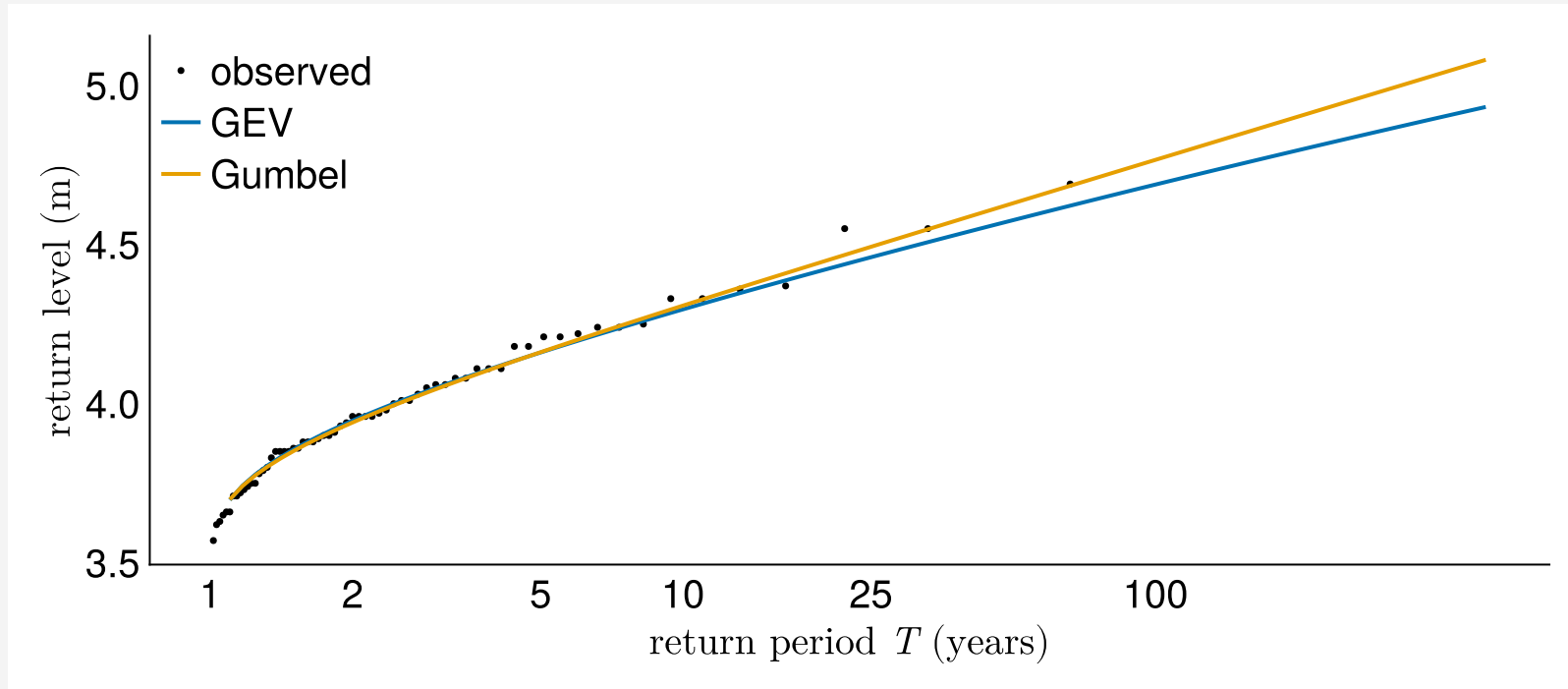
```
1 T_grid = exp.(range(log(1.1), log(500); length=100))
2 rl_gev = [quantile(d_gev, 1 - 1 / T) for T in T_grid]
3
4 function gumbel_nll(theta, y)
5     mu, logsigma = theta
6     sigma = exp(logsigma)
7     d = Gumbel(mu, sigma)
8     nll = 0.0
9     for y_i in y
10         lp = logpdf(d, y_i)
11         isfinite(lp) || return 1e10
12         nll -= lp
```

# DOES THE SHAPE PARAMETER MATTER?

```
13     end
14     return nll
15 end
16 res_gumbel = optimize(θ -> gumbel_nll(θ, y), [mean(y), log(std(y))], NelderMead())
17 d_gumbel = Gumbel(Optim.minimizer(res_gumbel)[1], exp(Optim.minimizer(res_gumbel)[2]))
18 rl_gumbel = [quantile(d_gumbel, 1 - 1 / T) for T in T_grid]
19 @printf("Gumbel MLE:  $\hat{\mu}$  = %.3f,  $\hat{\sigma}$  = %.3f", d_gumbel.μ, d_gumbel.θ)
```

Gumbel MLE:  $\hat{\mu}$  = 3.869,  $\hat{\sigma}$  = 0.195

# THE TWO CURVES DIVERGE IN THE TAIL



They agree at short return periods and diverge past 50 years.

# WHICH FIT SHOULD YOU USE?

$\hat{\xi} \approx -0.05$ : the GEV bends slightly downward, the Gumbel stays straight.

Coles (Coles, 2001, §3.5.1): the Gumbel's confidence interval on the 100-year level is narrower because forcing  $\xi = 0$  removes a source of uncertainty. The data do not rule out  $\xi \neq 0$ , so the GEV's wider interval is the one that reflects what the data actually say.

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# THE QQ PLOT

Sort the observed data from smallest to largest:  $y_{(1)} \leq y_{(2)} \leq \dots \leq y_{(n)}$ .  
These are the **observed quantiles**.

The **model quantiles** come from the fitted CDF. The  $i$ -th model quantile is the value the fitted distribution predicts the  $i$ -th smallest observation should take:

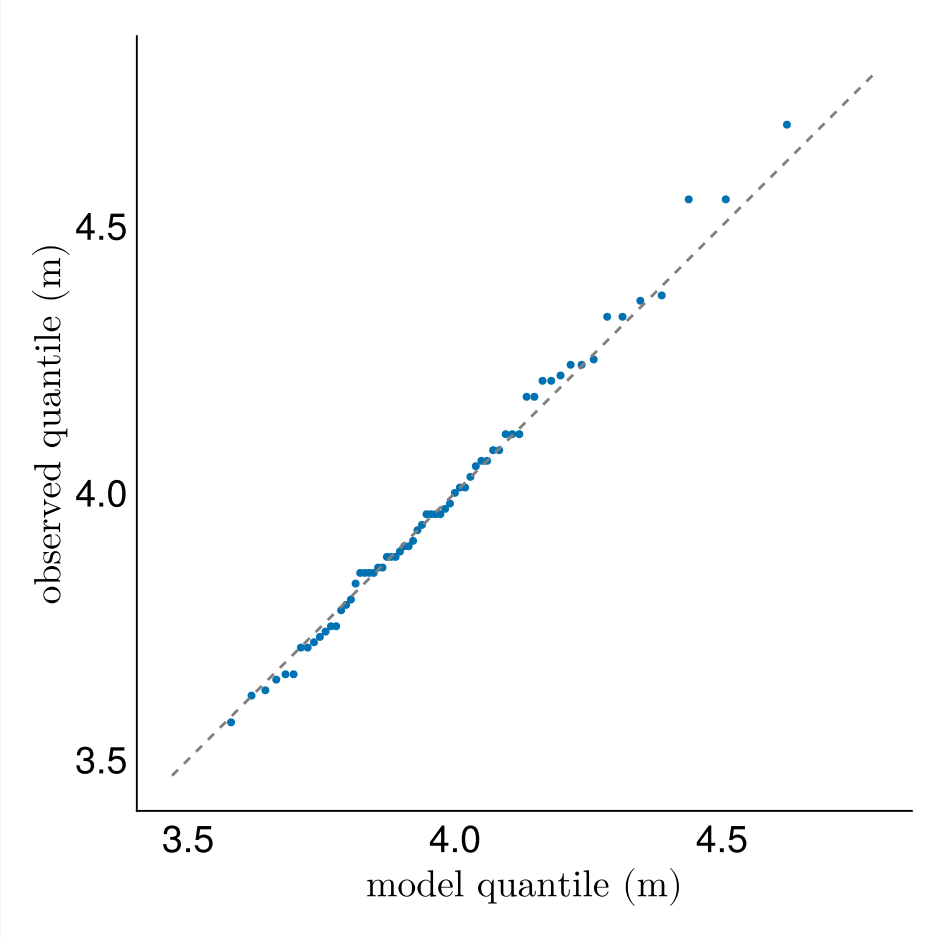
$$q_i = F^{-1}\left(\frac{i}{n+1}; \hat{\mu}, \hat{\sigma}, \hat{\xi}\right)$$

Plot one against the other. If the model fits well, each observed quantile is close to its model quantile, and the points fall near the diagonal.

# QQ PLOT FOR PORT PIRIE

```
1 n_obs = length(y)
2 y_sorted = sort(y)
3 model_q = [quantile(d_gev, i / (n_obs + 1)) for i in 1:n_obs]
4 fig = Figure(; size=(500, 500))
5 ax = Axis(fig[1, 1]; xlabel=L"\text{model quantile (m)}",
6           ylabel=L"\text{observed quantile (m)}", aspect=1)
7 scatter!(ax, model_q, y_sorted; color=colors[1], markersize=6)
8 lims = (minimum(y_sorted) - 0.1, maximum(y_sorted) + 0.1)
9 lines!(ax, collect(lims), collect(lims); color=:gray, linestyle=:dash)
10 fig
```

# QQ PLOT FOR PORT PIRIE

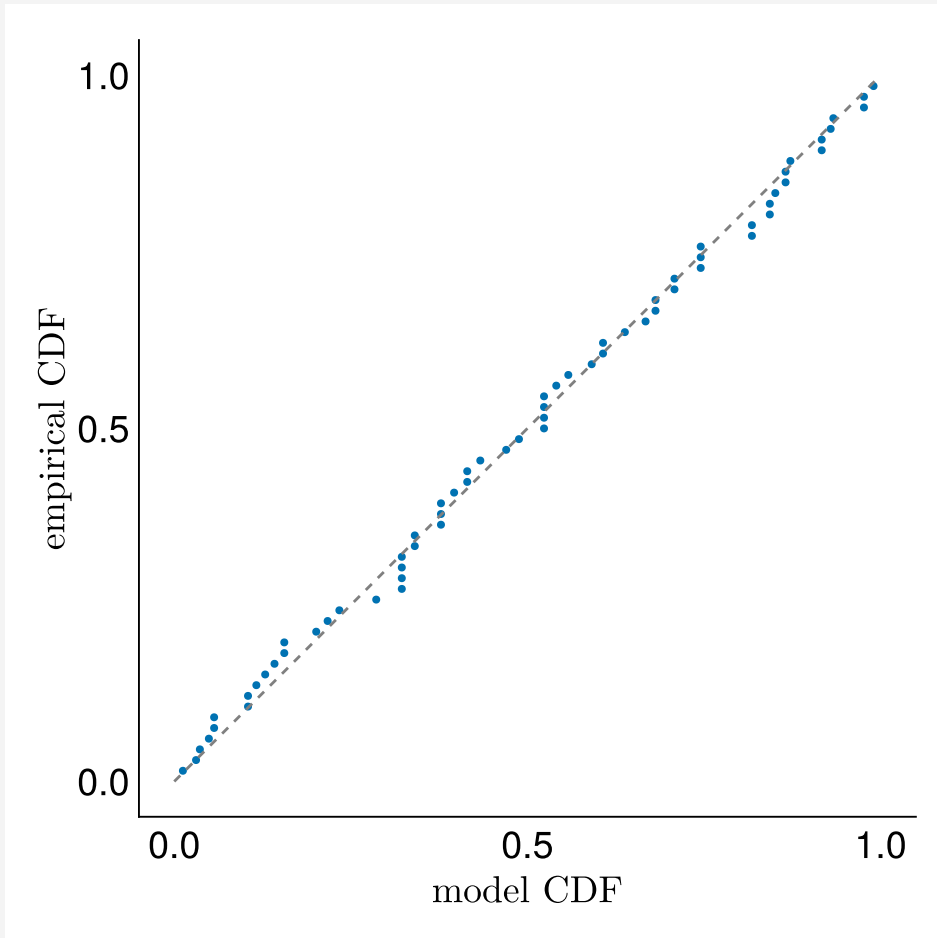


# THE PROBABILITY PLOT

Both axes on the CDF scale:  $\hat{F}$  vs  $F_{\hat{\theta}}$ . The probability plot checks the body of the distribution, where most of the data sit. It compresses the tail, so use the QQ plot for the upper end.

```
1 F_empirical = (1:n_obs) ./ (n_obs + 1)
2 F_model = [cdf(d_gev, y_sorted[i]) for i in 1:n_obs]
3 fig = Figure(; size=(500, 500))
4 ax = Axis(fig[1, 1]; xlabel=L"\text{model CDF}",
5           ylabel=L"\text{empirical CDF}", aspect=1)
6 scatter!(ax, F_model, F_empirical; color=colors[1], markersize=6)
7 lines!(ax, [0, 1], [0, 1]; color=:gray, linestyle=:dash)
8 fig
```

# THE PROBABILITY PLOT



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# WHY BAYES?

MLE gives one  $\hat{\theta}$  and one return-level curve. With 65 years of data, the parameters carry real uncertainty: a different 65-year draw from the same process would give a different  $\hat{\theta}$ . A Bayesian posterior is a distribution over  $\theta$  that reflects that uncertainty.

**Bayes' theorem:**

$$p(\theta \mid y) \propto p(y \mid \theta) p(\theta)$$

The posterior  $p(\theta \mid y)$  combines the likelihood with a prior. There is no closed-form expression for the posterior, but you can sample from it.

# MARKOV CHAIN MONTE CARLO

**Markov chain Monte Carlo** builds a chain of parameter vectors  $\theta^{(1)}, \theta^{(2)}, \dots$  whose distribution converges to the posterior.

The simplest version (Metropolis):

1. Propose  $\theta' = \theta^{(t)} + \varepsilon$ , where  $\varepsilon$  is a random step
2. Compute the acceptance ratio  $\alpha = p(\theta' | y) / p(\theta^{(t)} | y)$
3. Accept the proposal with probability  $\min(1, \alpha)$ ; otherwise stay at  $\theta^{(t)}$

Modern samplers (NUTS) use gradients to propose smarter steps, but the logic is the same: wander through parameter space, spending more time where the posterior is high.

# A GEV MODEL IN TURING

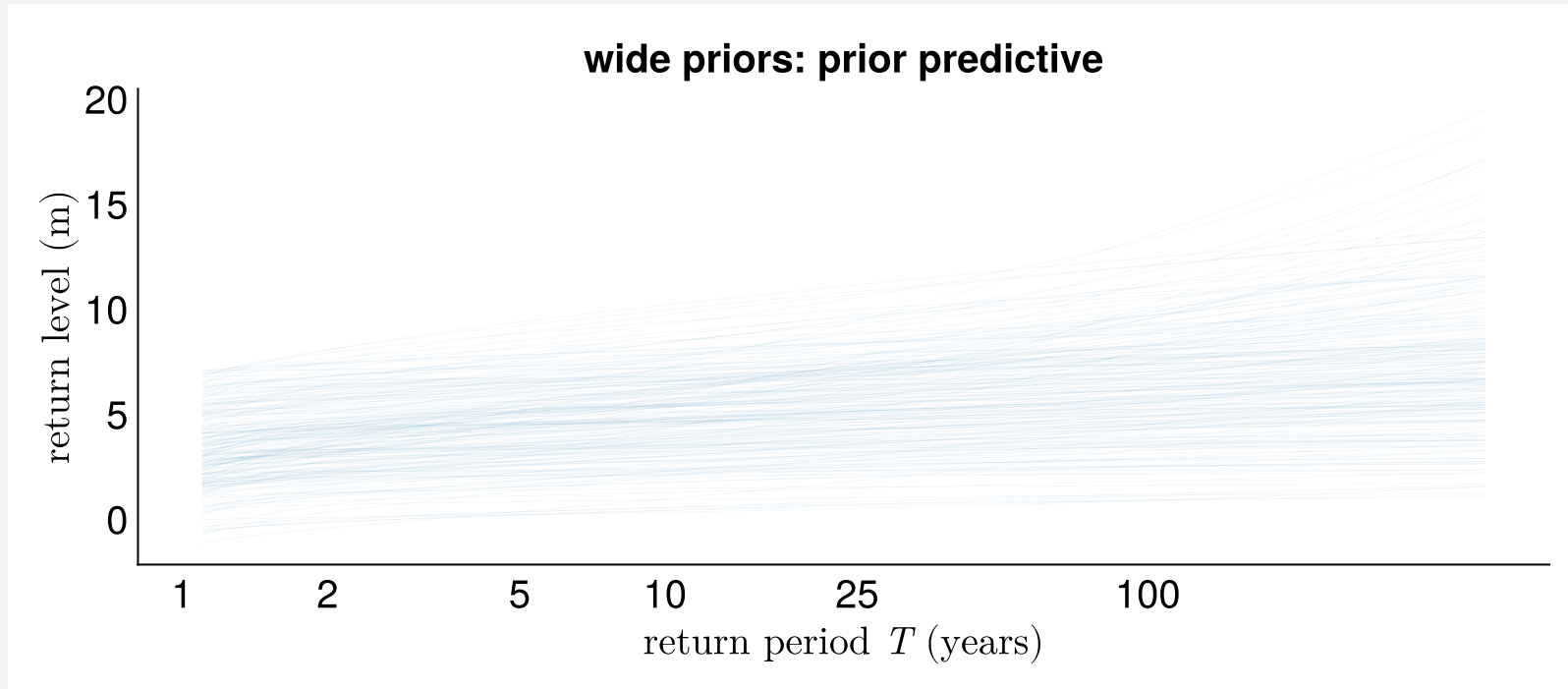
```
1 @model function gev_bayes(y)
2    $\mu$  ~ Normal(4.0, 2.0)
3    $\sigma$  ~ truncated(Normal(0.2, 0.5); lower=0.001)
4    $\xi$  ~ truncated(Normal(0, 0.3); lower=-0.35, upper=0.35)
5   if !any(ismissing.(y))
6     y .~ GeneralizedExtremeValue( $\mu$ ,  $\sigma$ ,  $\xi$ )
7   end
8 end
```

# PRIOR PREDICTIVE CHECK: ARE THE WIDE PRIORS REASONABLE?

A prior predictive check asks: do the priors, before seeing any data, produce return-level curves that look like something this site could plausibly produce? Pass `missing` as data so Turing samples everything from the prior.

```
1 n_miss = length(y)
2 wide_prior_chain = sample(gev_bayes(fill(missing, n_miss)), Prior(), 1000)
3 wide_df = DataFrame(wide_prior_chain)
```

# PRIOR PREDICTIVE CHECK: ARE THE WIDE PRIORS REASONABLE?



These priors place no constraint on the return levels: the 100-year sea level could be anything from  $-10,000$  to  $+10,000$  metres.

# QUANTILE PRIORS

Instead of reasoning about  $(\mu, \sigma, \xi)$  directly, express beliefs about observable quantities.

“The 2-year sea level at Port Pirie is around 4 m.” “The 100-year sea level is probably between 3 and 7 m.”

Translate each into a distribution over the return level at that return period, then add its log-density to the model (Coles, 2001, §9.5; Doss-Gollin & Keller, 2023).

# DEFINING QUANTILE PRIORS

```
1 quantile_priors = [  
2     (rt=2, dist=Normal(4.0, 1.0)),  
3     (rt=10, dist=Normal(4.5, 1.5)),  
4     (rt=100, dist=Normal(5.0, 2.0)),  
5     (rt=500, dist=Normal(6.0, 3.0)),  
6 ]
```

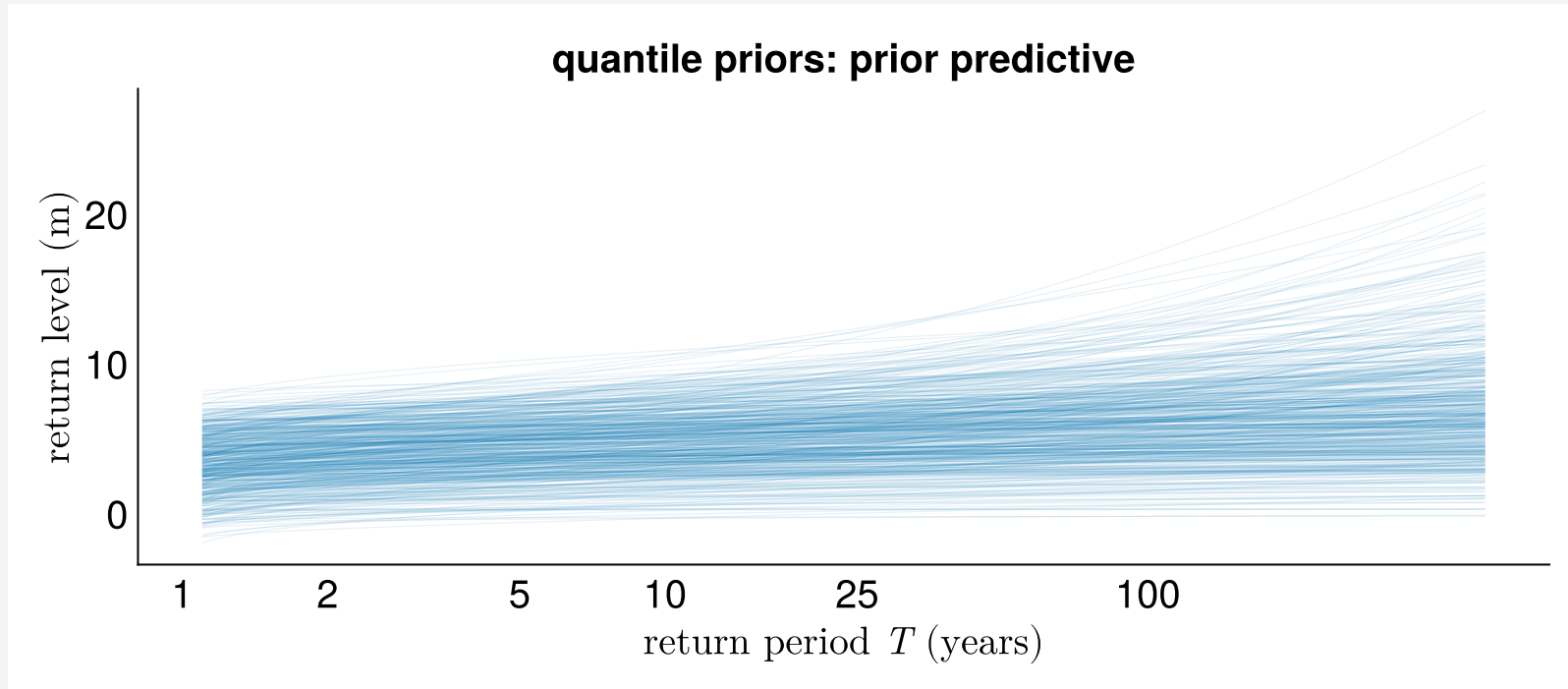
# THE MODEL WITH QUANTILE PRIORS

```
1 @model function gev_bayes_qp(y, priors)
2    $\mu$  ~ Normal(4.0, 2.0)
3    $\sigma$  ~ truncated(Normal(0.2, 0.5); lower=0.001)
4    $\xi$  ~ truncated(Normal(0, 0.3); lower=-0.35, upper=0.35)
5   dist = GeneralizedExtremeValue( $\mu$ ,  $\sigma$ ,  $\xi$ )
6   for p in priors
7     rl = quantile(dist, 1.0 - 1.0 / p.rt)
8     Turing.@addlogprob! logpdf(p.dist, rl)
9   end
10  if !any(ismissing.(y))
11    y .~ dist
12  end
13 end
```

# PRIOR PREDICTIVE CHECK: QUANTILE PRIORS

```
1 qp_prior_chain = sample(gev_bayes_qp(fill(missing, n_miss), quantile_priors), Prior(),  
2000)  
2 qp_df = DataFrame(qp_prior_chain)
```

# PRIOR PREDICTIVE CHECK: QUANTILE PRIORS



# PRIOR PREDICTIVE CHECK: QUANTILE PRIORS

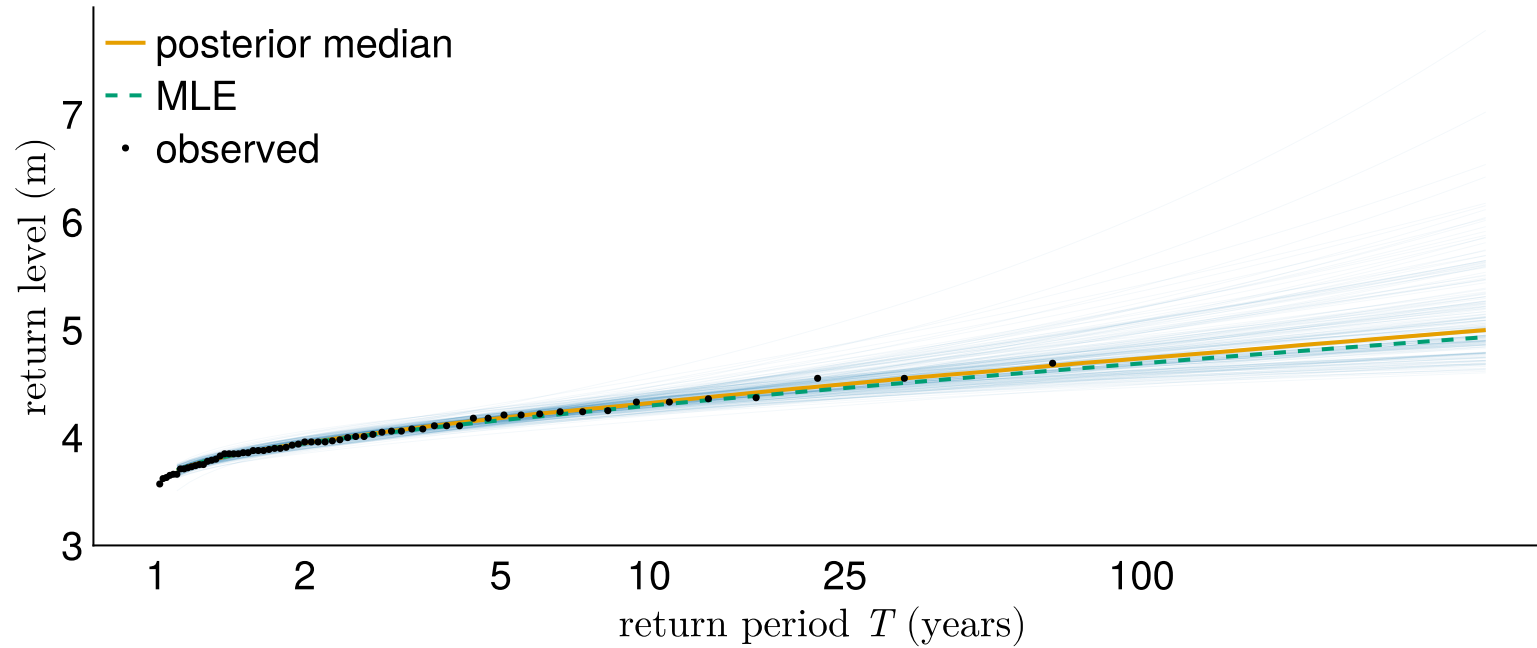
The quantile priors concentrate the curves toward physically plausible sea levels, but some draws still range widely. Compare this to the wide-prior plot: the quantile constraints are doing real work.

# POSTERIOR FIT

```
1 chain = sample(gev_bayes_qp(y, quantile_priors), NUTS(0.95), MCMCThreads(), 2_000, 4)
```

```
posterior draws: 8000
```

# POSTERIOR RETURN-LEVEL CURVES



# MLE VS BAYES: THE 100-YEAR LEVEL

```
MLE:    4.69 m  
Bayes:  4.74 m  [4.52, 5.33]  (95% credible)
```

The posterior median is close to the MLE. The credible interval accounts for parameter uncertainty, which the MLE point estimate alone does not.

# FRIDAY'S LAB

In the lab you fit a GEV to a NOAA tide gauge four ways:

- `gevfit` and `gevfit_lmom`, as above
- a log-likelihood you write and optimize yourself
- Turing, with wide priors and with quantile priors

Then you change the gauge and see which conclusions survive.